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1 Conclusion and References

Conclusion

1. **AI workloads expose parallel linear algebra.** Neural-network layers repeatedly reduce to matrix and tensor operations.
2. **GPUs already had the right foundation.** Graphics demanded parallel throughput and high memory bandwidth; unified shaders and CUDA made that hardware programmable.
3. **Data movement determines real performance.** Coalescing, shared-memory tiling, and register reuse move GEMM toward the compute limit.
4. **AI is now driving specialization.** Tensor Cores, lower precision, and new accelerators trade flexibility for throughput and efficiency.

Main takeaway

Graphics and AI both reward massive parallelism. The GPU succeeded by evolving from a graphics pipeline into a programmable matrix engine.

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Questions?